True bearings

1 State the true bearing of the following:

a Southwest

b Northwest

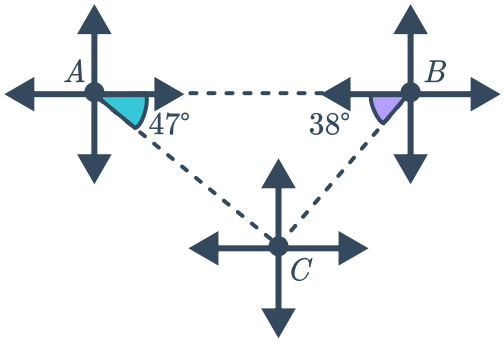
c West-Northwest

d South-Southeast

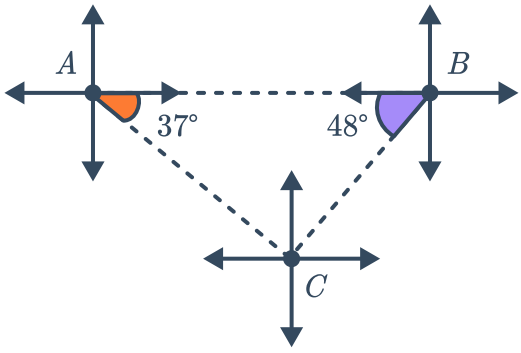
e Northeast

f Southeast

2 Determine the true bearing of point B from point A in the following bearings diagram:



3 Determine the true bearing of point A from point B in the following bearings diagram:



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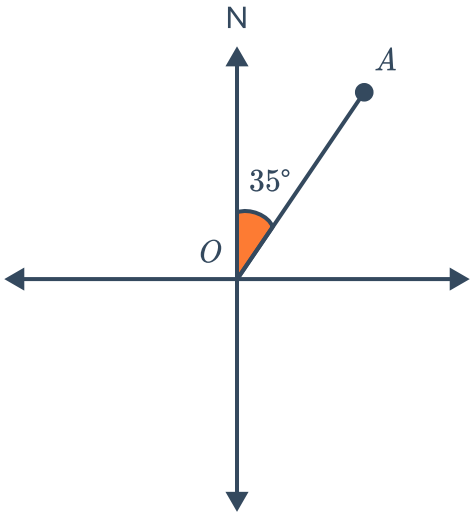
Compass bearings

4 Consider the point A in the folowing diagrams:

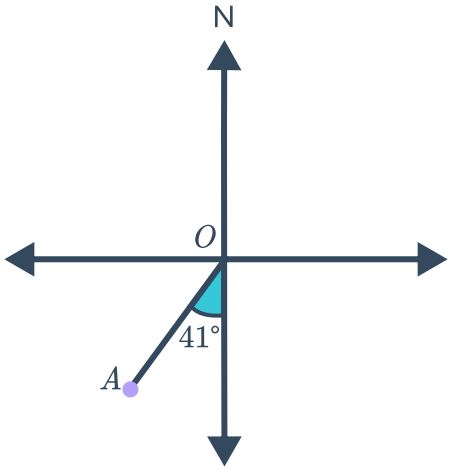
i Determine the true bearing of point A from O.

ii Find the compass bearing of point A from O.

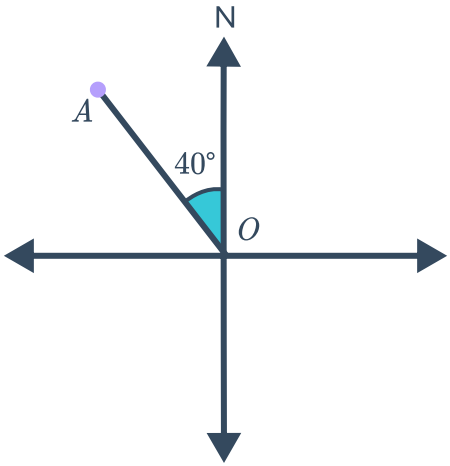
a



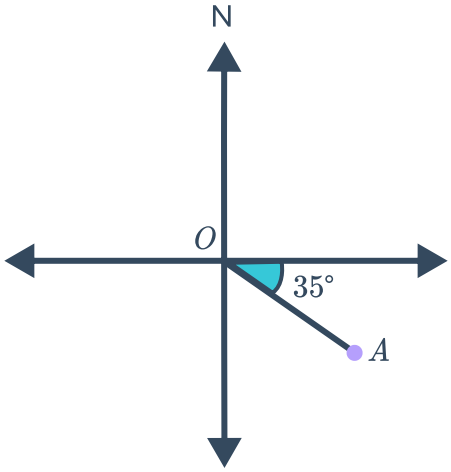
b



c



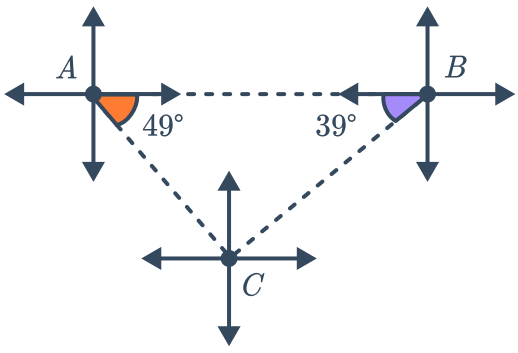
d



5 In the figure below, point B is due East of point A:

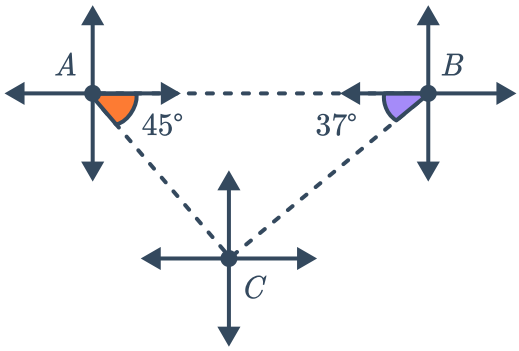
a Find the true bearing of point A from point C.

b Find the compass bearing of point A from point C.



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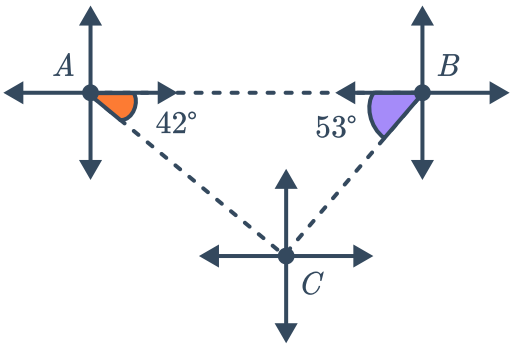
6 Consider the following diagram:



a Find the true bearing of point B from point C.

b Find the compass bearing of point B from point C.

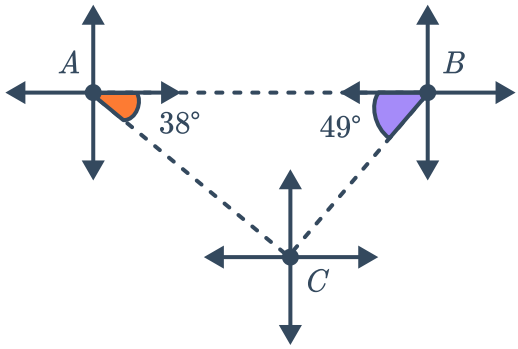
7 Consider the following diagram:



a Find the true bearing of point C from point A.

b Find the compass bearing of point C from point A.

8 Consider the following diagram:



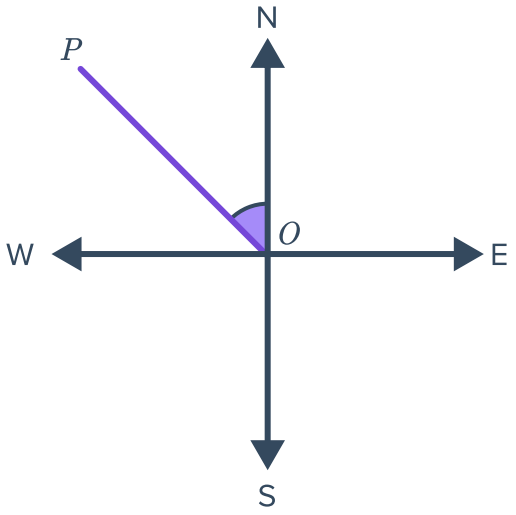
a Find the true bearing of point C from point B.

b Find the compass bearing of point C from point B.

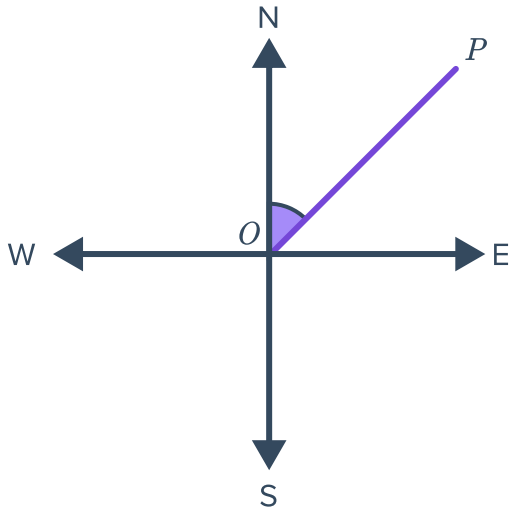
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9 Determine whether the following diagrams display the point P with a compass bearing between N and W from the point O.

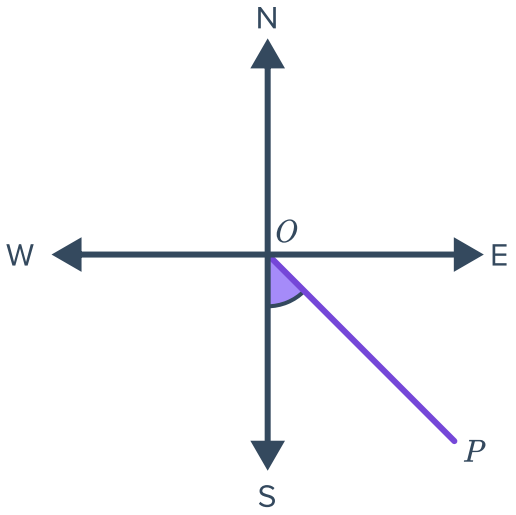
a



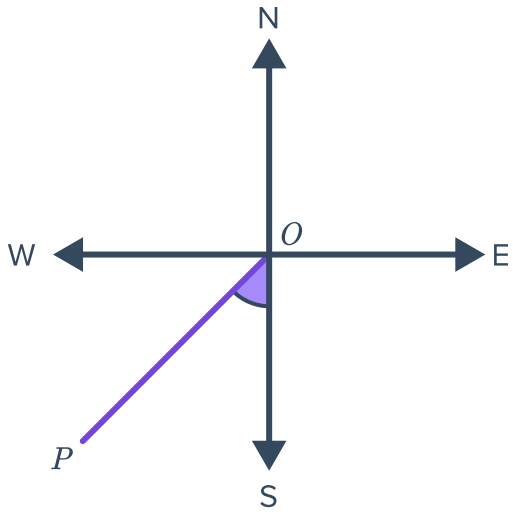
b



c



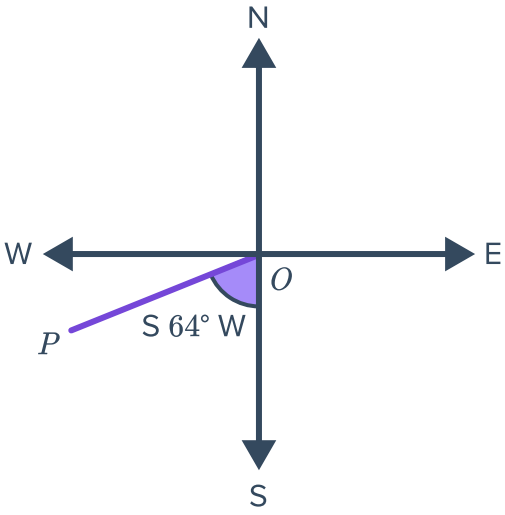
d



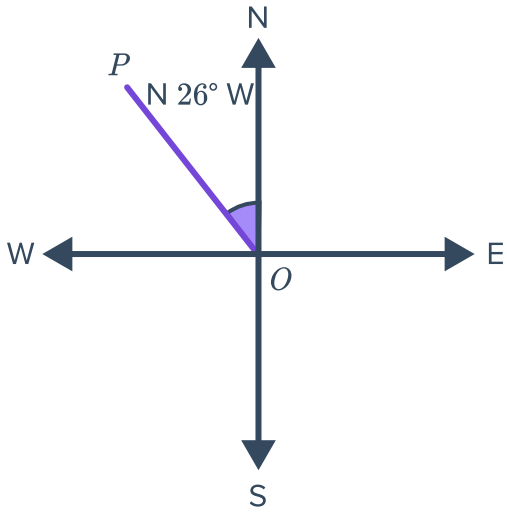
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10 Determine whether the following diagrams show a true bearing of 296° T for the point P from the origin O.

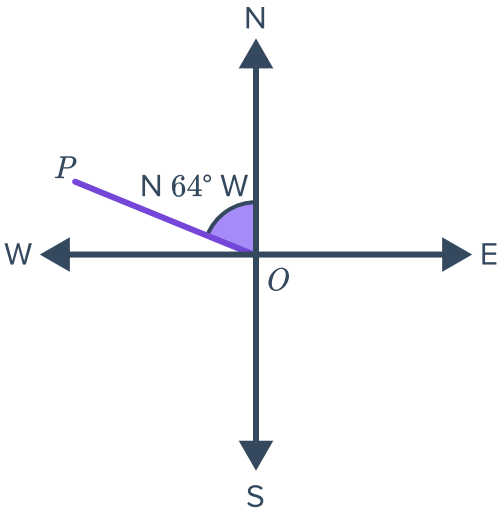
a



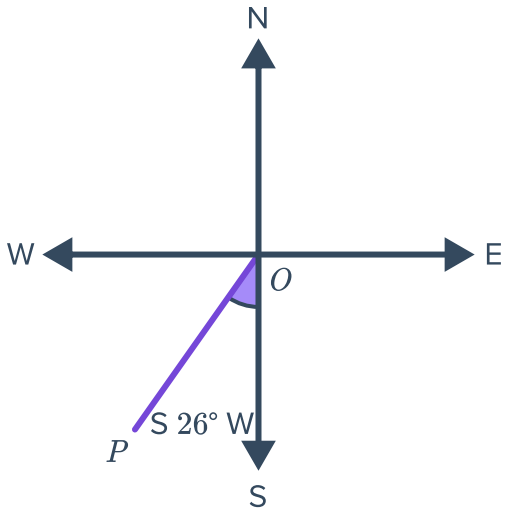
b



c



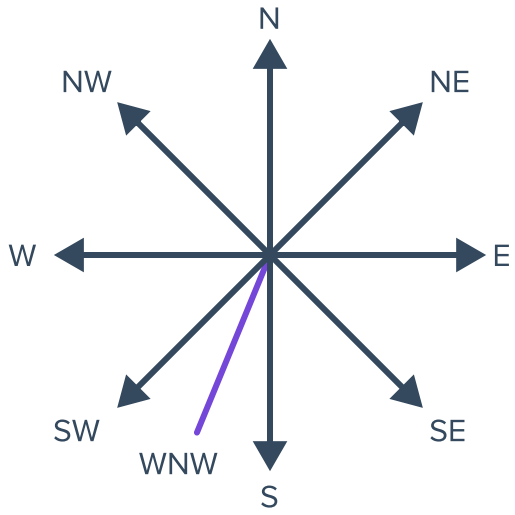
d



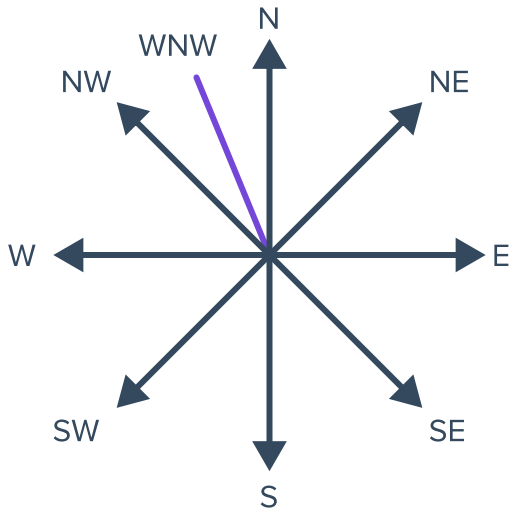
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11 Determine whether the following diagrams display WNW:

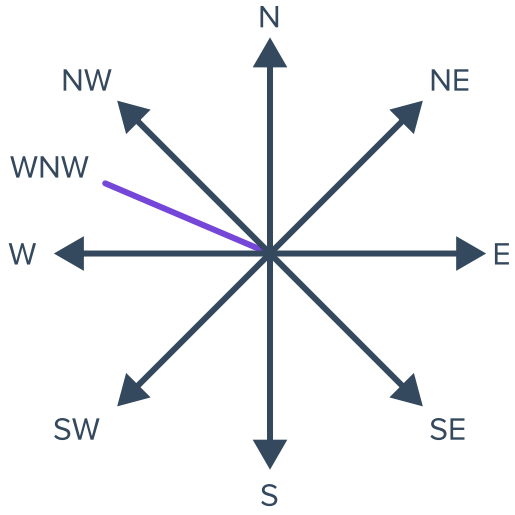
a



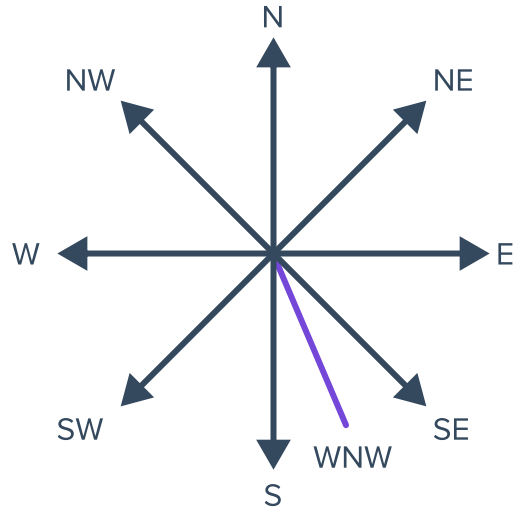
b



c



d

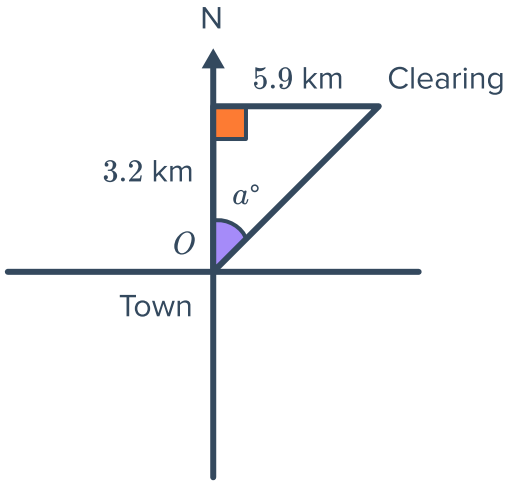


12 If a person walks at a compass bearing of SW, and then turns around to go back in the direction they came, what direction are they heading now?

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Applications

13 Consider the following diagram:
If the bearing of the clearing from the town is a°, find a to the nearest degree.

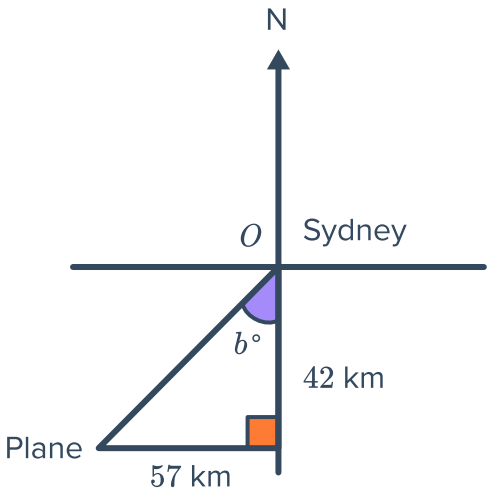


14 An aeroplane sets off on a bearing of 051°T. After some time, it needs to turn back and head to its original position. On what true bearing must it travel?

15 A boat sets off on a bearing of 125°T . After some time, it needs to turn back and head to its original position. On what true bearing must it travel?

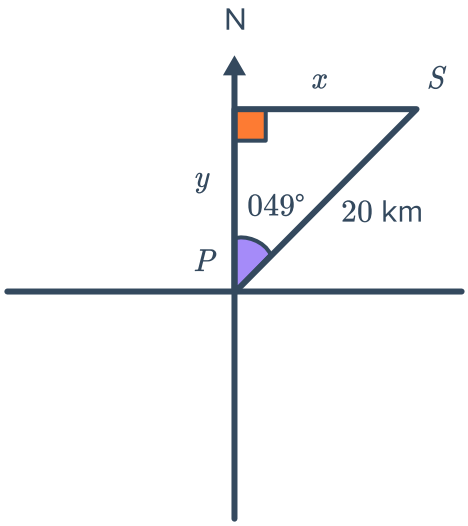
16 Shortly after take-off, a plane is $42\ \text{km}$ south and $57\ \text{km}$ west of the airport in Sydney that it left from:

- **a** Find the size of the angle marked b, to one decimal place.
- **b** Hence, find the compass bearing of the plane from the airport.
- **c** State the true bearing of the plane from the airport.



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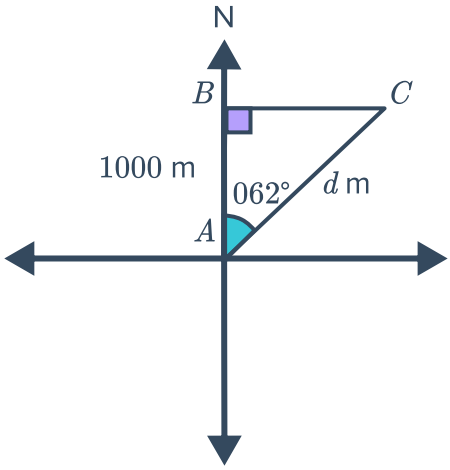
17 The position of a ship S is given to be $20\ \text{km}$ from P, on a true bearing of $049^\circ\ \text{T}$. The position of the ship can also be given by its (x, y) coordinates.



a Find the ship's x-coordinate to one decimal place.

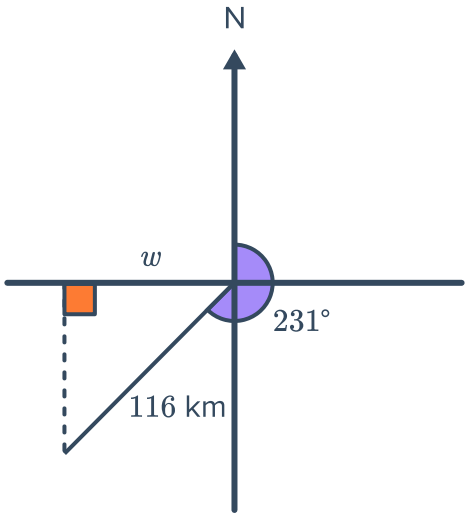
b Find the ship's y-coordinate to one decimal place.

18 On an orienteering course, Beth runs 1000 metres north from point A to point B, then turns east and runs to point C. If the true bearing of C from A is 062°T and the direct distance Beth must run to get back to point A is d metres, find d to the nearest metre.



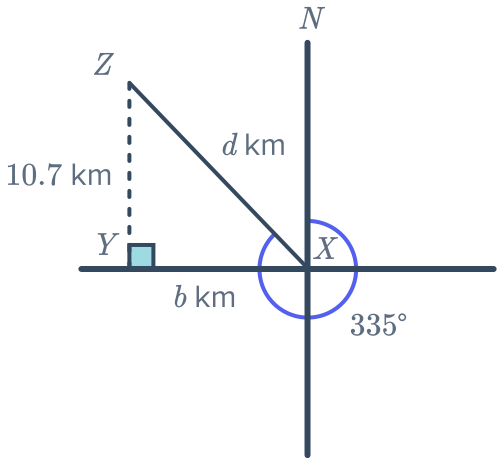
19 During a group hike, a hiker walked $11\ \text{km}\ \text{NE}$ of base camp. In the evening the hiker was separated from the group and after a night lost in the wilderness, the hiker was found $11\ \text{km}\ \text{NW}$ of the base camp. What compass bearing must the hiker have followed during the night?

20 Luke sailed for $116\ \text{km}$ on a bearing of 231°. If w is how many kilometres west he has sailed from his starting point, find w to one decimal place.



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21 During a rescue search, a helicopter flew west from point X to point Y, then changed course and flew 10.7 km north to point Z. If point Z is on a bearing of 335° T from point X:

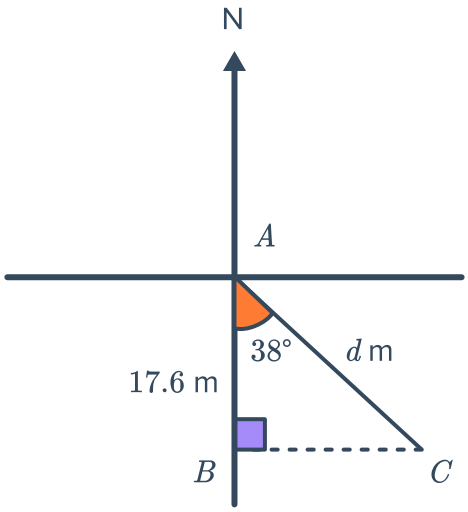


a Find the size of $\angle YXZ$.

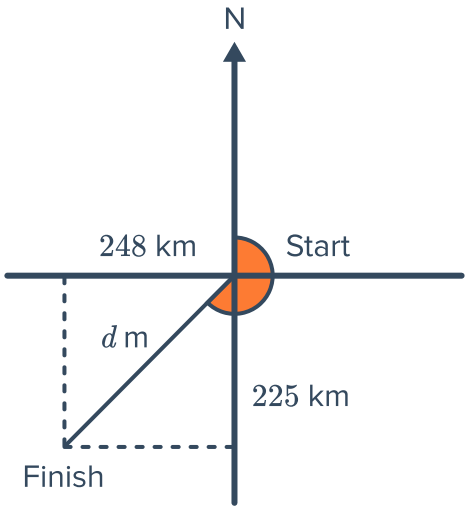
b If the distance from point Y to point X is b km, find b to one decimal place.

c If the distance that the helicopter must fly between point Z and point X is d km, calculate d to one decimal place.

22 Three television presenters are practising their navigation skills before heading off on an expedition to a remote location. Belinda at point B is positioned 17.6 m south of Amelia at point A. Carl at point C is due east of Belinda and on a bearing of S 38°E from Amelia. If Amelia and Carl are d m apart, find d to one decimal place.



23 A yacht sailed in a direction so that its final position was 248 km west and 225 km south of its starting point.



a If the true bearing on which the yacht sailed is b°, find the value of b to one decimal place.

b If the boat has sailed a total of d m, find the value of d to one decimal place.

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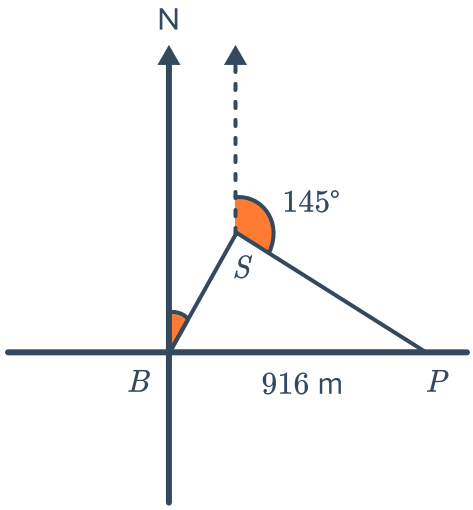
24 In remote locations, photographers must keep track of their position from their base. One morning a photographer sets out from base, represented by point B, to the edge of an ice shelf at point S on a bearing of 055°. She then walked on a bearing of 145° to point P, which is $916\ \text{m}$ due east of base.

a State the size of $\angle BSP$.

b Find the distance BS, to one decimal place.

c Find the distance SP, to one decimal place.

d If the photographer walks directly back to her base from point P, determine the total distance she would have travelled. Round your answer to one decimal place.



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